奕诚27单科卷-国网其他工学-数学

共 40 题（单选 / 多选 / 判断） 限时建议 90 分钟 卷末附答案速查与逐题解析

一、数学（40 题）

1.

设 $f(x)$ 可导，则 $\lim\limits_{\Delta x\to 0}\dfrac{f(x_0+m\Delta x)-f(x_0-n\Delta x)}{\Delta x}=$（　　）。

A. <img src="https://dati.gwzpks.com/uploads/20260723/tUxOe6IjFeAPptZqBQuDHq5sJMgCUGFfrUk5yB23.png">

B. <img src="https://dati.gwzpks.com/uploads/20260723/cpFpHWryhrX1LCUkHapS8dHg8C3IpoRe04fR2fEr.png">

C. <img src="https://dati.gwzpks.com/uploads/20260723/7SrzsVpZ9mcoMcINioGXcW1RWE6fmiNMIOjLgOHN.png">

D. <img src="https://dati.gwzpks.com/uploads/20260723/tc8X3J8dSG9N8ddOuaGlP55w3U74771pe3ketlvx.png">

2.

设 100 件产品中有 5 件是不合格品，今从中随机抽取 2 件，设 $A_1=\{$第一次抽的是不合格品$\}$，$A_2=\{$第二次抽的是不合格品$\}$，则下列叙述中错误的是（　　）。

A. <img src="https://dati.gwzpks.com/uploads/20260729/1XtUDhWrcSda6xvm3I7N99IIhXoCgtB6qsKZcJjH.png">

B. <img src="https://dati.gwzpks.com/uploads/20260729/jYkdPtrLM1laxBnwGaqM4ZTdydMrEiNw9KOV4kNC.png">

C. <img src="https://dati.gwzpks.com/uploads/20260729/QsRhIyKH8TqwYOOrxJfSw82U19SCpDYcVD1WHrNg.png">

D. <img src="https://dati.gwzpks.com/uploads/20260729/7xjlLftNCzVNcnJrYYW1H8fOIFIA2tWmo3Aa9Ftf.png">

3.

A 为三阶矩阵，$0,-1,1$ 为它的三个特征值，其对应的特征向量为 p_1,p_2,p_3。设 $P=[p_1 \quad p_2 \quad p_3]$，则下列等式正确的是（　　）。

A. <img src="https://dati.gwzpks.com/uploads/20260729/AcbsJQWfFWdebPQRghGXzHCe5wBGFEQg28fVu9WJ.png">

B. <img src="https://dati.gwzpks.com/uploads/20260729/mkHa6jTnHMoCcCdVghnA7QzgSzzO3oRqMb8vZC21.png">

C. <img src="https://dati.gwzpks.com/uploads/20260729/N4pxWsrbLNv7EmeTY26vpfygZstWM3SVosIHdzUX.png">

D. <img src="https://dati.gwzpks.com/uploads/20260729/HacG2FBbyvdxhwutxOx8PVyJ39VJAwdCuLKoHgc5.png">

4.

设 $X_1,X_2,\cdots,X_n$ 独立同分布，且 $E(X_i)=\mu$，则（　　）。

A. <img src="https://dati.gwzpks.com/uploads/20260729/rIdwhUlG4lDpkmmn43dfuCxsgntuumlu4FVOlp5u.png">

B. <img src="https://dati.gwzpks.com/uploads/20260729/k68s4RMbCLL6hiWd8pSZUdB6Otcov8Ab8FjTvPiD.png">

C. <img src="https://dati.gwzpks.com/uploads/20260729/AoYfApwcDwuXFQYIRcmrCWnRStmioGhcOaqFrEBd.png">

D. <img src="https://dati.gwzpks.com/uploads/20260729/Ir5JkZXKKa51dTDwtefPCAJpZeBxYiITr8nvMtrK.png">

5.

设 A 为 $m\times n$ 矩阵， C 为 n 阶可逆矩阵， $B=AC$ ，则（　　）。

A. <img src="https://dati.gwzpks.com/uploads/20260728/P2sXQPWisM1wv1lbm8G9QYlwrCWtzmeJaTSE8NqJ.png">

B. <img src="https://dati.gwzpks.com/uploads/20260728/N4trHhH6jcqkNg1RCVYJbACLfWBZRG7GlMF5myxW.png">

C. <img src="https://dati.gwzpks.com/uploads/20260728/soSn54UilmoOHVcKVDSpOaSjx3NdcC2IXcGQUnMw.png">

D. <img src="https://dati.gwzpks.com/uploads/20260728/SPFMPUv2wWYk2dIRu8phQVMf3lcQQoAjNmsibJCm.png">

6.

若 x_1,x_2 分别是方阵 A 的两个不同的特征值对应的特征向量，则 $k_1x_1+k_2x_2$ 也是 A 的特征向量的充分条件是（　　）。

A. <img src="https://dati.gwzpks.com/uploads/20260729/CoFGlRUix54znohsUZZ1Tdu41BNl299JPtOWibMn.png">

B. <img src="https://dati.gwzpks.com/uploads/20260729/KFOo33c3SjhuGg5kcOO5uuJWbenulpGqVwSgaYWO.png">

C. <img src="https://dati.gwzpks.com/uploads/20260729/xIuMNPm3PBWlhHLTgHufqT3f6eLseVOXVB3EFRgl.png">

D. <img src="https://dati.gwzpks.com/uploads/20260729/dUVP19CD3ET8CEb7seAoLVgf1b6Ak4sQJx9sVEsb.png">

7.

设 Ω 由 $z=x^2+y^2$ 与 $z=\sqrt{2-x^2-y^2}$ 围成，则 $\iiint\limits_{\Omega}z\mathrm{d}v=$（　　）。

A. 0　　B. 1

C. <img src="https://dati.gwzpks.com/uploads/20260723/8ViwbUxDsWQQtPzUVoGuL46jTXiEsyNrzfjlYEaZ.png">

D. <img src="https://dati.gwzpks.com/uploads/20260723/NnfXeI3TqCgVpukN2HOMtLSXVsqT1wyYk8B3vFJe.png">

8.

$\lim\limits_{x\to\infty} x(\sqrt{x^2+1}-x)=-\dfrac{1}{2}$。（　　）

A. 正确　　B. 错误

9.

设η_n表示n次独立重复试验中事件A出现的次数，p是事件A在每次试验中出现的概率，

则$P\{a<\eta_n\le b\}\approx \int_{\frac{a-np}{\sqrt{np(1-p)}}}^{\frac{b-np}{\sqrt{np(1-p)}}}\frac{1}{\sqrt{2\pi}}e^{-\frac{t^2}{2}}dt$。（　　）

A. 正确　　B. 错误

10.

设向量组A：$\boldsymbol{\alpha}_1=(a_{11},a_{21},a_{31})^{\mathrm{T}},\boldsymbol{\alpha}_2=(a_{12},a_{22},a_{32})^{\mathrm{T}},\boldsymbol{\alpha}_3=(a_{13},a_{23},a_{33})^{\mathrm{T}}$，向量组$B$：$\boldsymbol{\beta}_1=(a_{11},a_{21},a_{31},a_{41})^{\mathrm{T}},\boldsymbol{\beta}_2=(a_{12},a_{22},a_{32},a_{42})^{\mathrm{T}},\boldsymbol{\beta}_3=(a_{13},a_{23},a_{33},a_{43})^{\mathrm{T}}$，则（　　）。

A.
src="https://dati.gwzpks.com/uploads/20260728/eCsrtU7EQv8cMfRo8wsu58slnDl4rGAawZIYLowT.png">

B.
src="https://dati.gwzpks.com/uploads/20260728/YF7DlyhoJPKwHpeVgyUyem8QljAwujPNFOts8TPQ.png">

C.
src="https://dati.gwzpks.com/uploads/20260728/93N3llMIk9PmmUepo6upmgarFLgAizjfod5PPRkU.png">

D.
src="https://dati.gwzpks.com/uploads/20260728/YJXeVkD8TIfSAz2pIDn5K98oRc1z3TOImqYgwFju.png">

11.

设随机变量X和Y的数学期望分别为-2和3，方差分别为1和4，而相关系数为-0.5，则根据切比雪夫不等式，$P\{|X+Y|\geq 6\}$（　　）。

A. <img src="https://dati.gwzpks.com/uploads/20260729/naZaE7JbAQVHeRa5H2Kl0nvTIyqE5aXd2d5YdlsS.png">

B. <img src="https://dati.gwzpks.com/uploads/20260729/RxSxcSkiLbJd6mZYVRN0fwpBodhVoXvsiF6QWohj.png">

C. <img src="https://dati.gwzpks.com/uploads/20260729/SlgEq5Ti9NBBh5lIAOiKXs080te3LTnDJwM5JRwm.png">

D. <img src="https://dati.gwzpks.com/uploads/20260729/L3Is70wl26EM8CMeSBMOePqcDdWKaoirHV3Xr9a4.png">

12.

下列矩阵中是正定矩阵的为（ ）。

A. <img src="https://dati.gwzpks.com/uploads/20260729/Eh1MtpKgsE8W3uuuY8srMSe3ty3IT9IbQxzpB3F8.png">

B. <img src="https://dati.gwzpks.com/uploads/20260729/grmxKWqbyGX69kGS0pc15zagxCYL6KbqbHdnRa5B.png">

C. <img src="https://dati.gwzpks.com/uploads/20260729/n2qPsBmBCTrxTLYLA1FizZx7WcpbnFpL3OzJvQ4N.png">

D. <img src="https://dati.gwzpks.com/uploads/20260729/8AtauJsMOHaWkNqwc55N3xPdLpFoC1Hlv2BnFhI8.png">

13.

曲线$\begin{cases} x=e^t\sin 2t \\ y=e^t\cos t \end{cases}$在点$(0，1)$处的法线方程为$y-1=-2(x-0)$。（　　）

A. 正确　　B. 错误

14. 

A. <img src="https://dati.gwzpks.com/uploads/20260729/ZAPlkyvbHXF4oOng010Vvuc1eaFGPqJWRvQuxUId.png">

B. <img src="https://dati.gwzpks.com/uploads/20260729/UAVlUWGQC3rcvObsyJjr9dZM6TyCNkwjPuheCSM5.png">

C. <img src="https://dati.gwzpks.com/uploads/20260729/uZqvPoY3i8kbIYQvvhiydrtl5BgwWmtozQSB7Twp.png">

D. <img src="https://dati.gwzpks.com/uploads/20260729/eWtNAn3FhxFk0qtqCwOmMoK2dsxIe3kGapYYT6Vf.png">

15. 

A. -3　　B. -1

C. 1　　D. 3

16. 

A. <img src="https://dati.gwzpks.com/uploads/20260723/mCPsR3A42fUT54vMOUCRuhjYlEp7RsFngTtBTBk7.png">

B. <img src="https://dati.gwzpks.com/uploads/20260723/qAgPAtXoOOe5L6YZplmxgtXmOCPW1xH6gnnlHy1s.png">

C. b=1　　D. b=2

17. 设A为可逆矩阵，则与A一定有相同特征向量的矩阵为（ ）。

A. <img src="https://dati.gwzpks.com/uploads/20260728/gKQB96FTSyJB6NAWmYWxvDwQXJSJUodnOEM3HiYl.png">

B. <img src="https://dati.gwzpks.com/uploads/20260728/AbNB9wmEmWOwnQjrNlBOPs5fnkkZX4jr6u8od8gw.png">

C. <img src="https://dati.gwzpks.com/uploads/20260728/AL2zFDmvhEcj2rPRdInybcoVYKPPo9SJSNnYVrCT.png">

D. <img src="https://dati.gwzpks.com/uploads/20260728/wzN4cBPSrGgxgMDbTi1eZO2jtAe7UqvMKo1I3VZX.png">

18. 

A. -1　　B. -2

C. -3　　D. 0

19.

A. A,B互斥　　B. A,B独立

C. <img src="https://dati.gwzpks.com/uploads/20260729/wnMt5IHp5Oa9MnDY93K6gUKpibSAZpkqERpWJUJH.png">

D. <img src="https://dati.gwzpks.com/uploads/20260729/KuSGjuAWPZOtNydxzGtNMe8Gu6SzfMhJrHjDBzDm.png">

20.

A. <img src="https://dati.gwzpks.com/uploads/20260729/sxXqoEfkm0xDmUes1oiD1ppM2HbmBWuGiBh8Zuq0.png">

B. <img src="https://dati.gwzpks.com/uploads/20260729/yd3VteDCCr007nZKqVggadWxu4xTy91Rq8i49DK7.png">

C. <img src="https://dati.gwzpks.com/uploads/20260729/RjGNEa5vTS1MfyoDZPoZtYvk6xNrU256GGM17rn0.png">

D. <img src="https://dati.gwzpks.com/uploads/20260729/r8CxN7epjFDz0mL7kKeP2ToXx0vZZFBYh8GKZ74a.png">

21.

A. 每一个向量都能由其余三个向量线性表示 B. 只有一个向量能由其余三个向量线性表示

C. 只有一个向量不能由其余三个向量线性表示 D. 每一个向量都不能能由其余三个向量线性表示

22.

A. <img src="https://dati.gwzpks.com/uploads/20260729/I8qn0YYZUJz0EsC4roT4DWMXX15007ZvuzCns90X.png">

B. <img src="https://dati.gwzpks.com/uploads/20260729/S3Na0x8JBow63VhQeNr40Uw6WXwIAGigExH3U332.png">

C. <img src="https://dati.gwzpks.com/uploads/20260729/mp4s2DhStkC6LbDIzSSNw4i0c1UyKC4WyVH5dyE4.png">

D. <img src="https://dati.gwzpks.com/uploads/20260729/QI0wZFIhwvREaXHCKwWMquL9CecB8zAehWe0P6Gg.png">

23.



A. 一定连续　　B. 不连续

C. 不一定连续　　D. 无法判断

24.



A. 没有间断点　　B. 有一个第一类间断点

C. 有两个以上第一类间断点　　D. 有一个间断点，但类型不确定

25.



A. <img src="https://dati.gwzpks.com/uploads/20260725/GarVjl7OP7P8ClRpsX219cjuLKgp1DjByOVOB5VP.png">

B. <img src="https://dati.gwzpks.com/uploads/20260725/gQuXC55dPJAGqVrO4beGXMlHKDj7fBWaqKwsMVt5.png">

C. <img src="https://dati.gwzpks.com/uploads/20260725/vLOGsGtFelQI8rOebmnxTk3G1mHlOyp4a6hPxkGn.png">

D. <img src="https://dati.gwzpks.com/uploads/20260725/VlenitxMUG5umUcz1NCK542JaaogAHYSTJZIZyOR.png">

26.



A. 3或4　　B. 1或2

C. 1或3　　D. 2或4

27.



A. <img src="https://dati.gwzpks.com/uploads/20260728/hxDL6l2lyE5GQik2UIudhV3emAqOP6KeagmlYcw9.png">

B. <img src="https://dati.gwzpks.com/uploads/20260728/itzGjbmWZOYTbVmooKRJMm62pwsQGxClXU9aVkX6.png">

C. <img src="https://dati.gwzpks.com/uploads/20260728/bkFnhKfgl9Y5KfyNOx5GauA2m31K2JaZ3gOn89wE.png">

D. <img src="https://dati.gwzpks.com/uploads/20260728/VQHgNjz1SJT8h2F2oJwKj4930U3xRjWtVoZlpGq4.png">

28.



A. 正确　　B. 错误

29.

A. 0.1　　B. 0.25

C. 0.5　　D. 0.7

30.

A. <img src="https://dati.gwzpks.com/uploads/20260725/31PAf50LJghg4QWnLwprVxomS0rw8E0LmHzoGtG6.png">

B. <img src="https://dati.gwzpks.com/uploads/20260725/gUKmfsy6V124NtFGhfBHNWHjlaeiJPPYa2SXIvk0.png">

C. <img src="https://dati.gwzpks.com/uploads/20260725/XSDsAZEYiMPQMfyROckdDp88DXjvXeG9a7JIzERe.png">

D. <img src="https://dati.gwzpks.com/uploads/20260725/zMBjuKDibyvDFb9A3XkyIRO9VuHneYzJ98ZPEQ7Z.png">

31.

A. <img src="https://dati.gwzpks.com/uploads/20260723/UzQzReqwBtoZ9UecO2UhmlsfHmWq9LzPlgyoOLT9.png">

B. <img src="https://dati.gwzpks.com/uploads/20260723/XoHtiJjHmfDTLA56GO92jGCZV5FhU6vfKZUkZgzt.png">

C. <img src="https://dati.gwzpks.com/uploads/20260723/UCMhETDkpsYcMuXJTMmhSB5sEEVVwHdhpUPAGCQ2.png">

D. <img src="https://dati.gwzpks.com/uploads/20260723/ZegBMih9NquEGweu1hj798MdUm4qSY3QQMYXfAnU.png">

32.



A. <img src="https://dati.gwzpks.com/uploads/20260729/2cUkDlSNEGBNoh9lD3pxv1gw3Yu9auEyxtcL6diI.png">

B. <img src="https://dati.gwzpks.com/uploads/20260729/C67Xk05F6PBgxSyMNk2ZfTNxuqCeh0itIZz8ck0T.png">

C. <img src="https://dati.gwzpks.com/uploads/20260729/h6jhWgijwakW85R8OYGRFS03Hd9mY6ynGuTf80ag.png">

D. <img src="https://dati.gwzpks.com/uploads/20260729/u1PspXVm3E3083nAQTQ82MBm1o86CUyFCjwRfAFu.png">

33.



A. <img src="https://dati.gwzpks.com/uploads/20260723/P7cnuUQPTW1h0MDvgUVWNIynVrZKRxJSWHmtoFUR.png">

B. <img src="https://dati.gwzpks.com/uploads/20260723/SyOgRxdBChLr5K4h2QtmE1pKYh2De4sTfHPtgAmT.png">

C. 函数沿任一方向的方向导数都存在　　D. 函数不一定连续

34.



A. <img src="https://dati.gwzpks.com/uploads/20260729/TRwbV1oXaYwD8i2az14GyWhUa5XLndySEVVM3VFw.png">

B. <img src="https://dati.gwzpks.com/uploads/20260729/PrBI9pZmcLesempBDocQJ1ZQWdNJ6fWC1mai0uno.png">

C. <img src="https://dati.gwzpks.com/uploads/20260729/lNaVwToZTzK9917cqN8AgDo2u69hqZkl9OEnV1xD.png">

D. <img src="https://dati.gwzpks.com/uploads/20260729/ry8ADiP5DOQk2DHqSXD6Ek64E30qwnVWF1ELmhAY.png">

35.



A. 正确　　B. 错误

36.



A. [-2,0]　　B. （-2,0）

C. (-2,0]　　D. [-2,0)

37.



A. <img src="https://dati.gwzpks.com/uploads/20260723/yzvvYObWriLZ3UHVcNNM4HRbuFbqruBLHV1aXMBC.png">

B. 0　　C. 1

D. 2

38.



A. <img src="https://dati.gwzpks.com/uploads/20260729/9Ya5FmM0ISej04gCnAsKR5mqEzryX8T9Z40jHdPe.png">

B. <img src="https://dati.gwzpks.com/uploads/20260729/dMNUaJkGB7pLkj3scZ5VLivXzZbbJWDCNi3Qwnq6.png">

C. <img src="https://dati.gwzpks.com/uploads/20260729/U4CLSWDHG2tqiUR7U37pfcFWYKR7sGYxjwNr38m8.png">

D. <img src="https://dati.gwzpks.com/uploads/20260729/btyo7F0sfSYme1sNnem9EeseguILh1xANa2JPFxj.png">

39.



A. <img src="https://dati.gwzpks.com/uploads/20260729/8sXKoVqJFQN4FRHyXrdM3GKWQLWoPDDAq1bqIoVL.png">

B. <img src="https://dati.gwzpks.com/uploads/20260729/qnknBApahSE5NNOQcGgSXaFVsTjTrmuZbWUosiAl.png">

C. <img src="https://dati.gwzpks.com/uploads/20260729/p7mqQXsJj7UHYn12czMxZJBlwFNfbfkN1jC0urRc.png">

D. <img src="https://dati.gwzpks.com/uploads/20260729/enkfILOkotEgyb7ItDK8TpkA05aICmhiNeQucozS.png">

40.



A. 正确　　B. 错误

答案速查

奕诚27单科卷-国网其他工学-数学

1.A	2.D	3.ABD	4.AD	5.BD	6.D
7.D	8.B	9.A	10.B	11.A	12.AC
13.A	14.B	15.D	16.AC	17.ABC	18.B
19.D	20.C	21.A	22.A	23.A	24.B
25.B	26.B	27.BCD	28.B	29.B	30.A
31.D	32.A	33.AC	34.AD	35.A	36.C
37.A	38.AD	39.A	40.A		

逐题解析

奕诚27单科卷-国网其他工学-数学

1. 答案 A

2. 答案 D

3. 答案 ABD

4. 答案 AD

5. 答案 BD

6. 答案 D

7. 答案 D

8. 答案 B

9. 答案 A

10. 答案 B

11. 答案 A

12. 答案 AC 【定位】本题考查10章，2节，知识点3。 【难度】中等+计算 【解析】顺序主子式大于0，B不满足实对称矩阵，D不满足二阶顺序主子式大于0。

13. 答案 A

14. 答案 B

15. 答案 D

16. 答案 AC

17. 答案 ABC

18. 答案 B

19. 答案 D

20. 答案 C

21. 答案 A

22. 答案 A <img src="https://dati.gwzpks.com/uploads/20260729/h9ttkrSzFcdoFRdHBP66tZQDcmfMj6dLXGAsxsYb.png">

23. 答案 A <img src="https://dati.gwzpks.com/uploads/20260723/t76ct6G11tY7nP74h2eGUPoBNEdONJdU67vKQhLl.png">

24. 答案 B <img src="https://dati.gwzpks.com/uploads/20260723/2UfAOFQVHOu6F1639yyAi4RrfNvRhJC64rFVJ2qe.png">

25. 答案 B <img src="https://dati.gwzpks.com/uploads/20260725/v9T6pSQ4BpmSJRcpywaMFxoMT4TtJLKreDxuGLgl.png">

26. 答案 B <img src="https://dati.gwzpks.com/uploads/20260728/DbVHw2vNcGaZXz2elXjzIXBFZgWCsUP92NA6j3iG.png">

27. 答案 BCD 【定位】本题考查8章，1节，知识点2。 【难度】简单+计算 【解析】方程组的解的结构

28. 答案 B <img src="https://dati.gwzpks.com/uploads/20260723/JOVgAyimzBb4NaYsfSPfJeKNEWAjxLvMQtvEh1kj.png">

29. 答案 B <img src="https://dati.gwzpks.com/uploads/20260729/6OD9TduQOgyNQGBYunm2E2jHbNXYaLXYTcFBud5w.png">

30. 答案 A <img src="https://dati.gwzpks.com/uploads/20260725/3GtgiS5VhoeyxiugjVwb3c79MNeGEJFxZs6E3QIv.png">

31. 答案 D <img src="https://dati.gwzpks.com/uploads/20260723/YhBnvVH7v3KcanVlQEoiaVPF97ePLHOwFhSzOOg1.png">

32. 答案 A <img src="https://dati.gwzpks.com/uploads/20260729/XQMDJKglkzqosUWF8tMVRy81yA8bNNYzUoX1GqoF.png">

33. 答案 AC 【定位】本题考查4章，3节，知识点4。 【难度】简单+计算 【解析】可微必可偏导，函数沿任一方向的方向导数都存在。

34. 答案 AD <img src="https://dati.gwzpks.com/uploads/20260729/HIYnll1i05ymErcOMe8INCiNcgCkeKnv4goeayiY.png">

35. 答案 A <img src="https://dati.gwzpks.com/uploads/20260729/hDL6gV2weaOd5m5jo6B5by8bTEPO4GbgwU29TkAO.png">

36. 答案 C <img src="https://dati.gwzpks.com/uploads/20260725/PxdSyODPH44PBzc2ghOtlZpmCtzakQkSCxEwlY1B.png">

37. 答案 A <img src="https://dati.gwzpks.com/uploads/20260723/8CglR7l7ICouGAzTV7QRckfZePCWJskwqo6Bt8Z7.png">

38. 答案 AD <img src="https://dati.gwzpks.com/uploads/20260729/d3oqZyspKcHxYFTvq3xuTq50cg2GB2D5yqwwF9ym.png">

39. 答案 A <img src="https://dati.gwzpks.com/uploads/20260729/5uz1uXWrbYQJQfhzQE8pMOnbqo8jJt7pJCAVRF1N.png">

40. 答案 A <img src="https://dati.gwzpks.com/uploads/20260911/rL9SmGdBcRm963fQdEMxytHeWhO82eDmqPuMfpsp.png">